Bike Sharing System Analysis

Project Overview

- This project analyzes a Bike Sharing System dataset to evaluate bike station capacity, bike availability, and service features across multiple cities.
- The Power BI dashboard provides insights into station capacity distribution, bike availability levels, geographic distribution, and time-based trends to understand how efficiently the bike-sharing system operates.

Objective

- Analyze total bike stands, available bikes, and available bike stands.
- Understand city-wise distribution of bike stations.
- Identify stations offering banking and bonus facilities.
- Analyze average bike availability and stand availability across cities.
- Track bike availability trends over time.

Problem Statement

- Uneven distribution of bike stands across cities may affect accessibility for users.
- Some stations experience low or zero available bikes, creating difficulty for users to rent bikes.
- Certain stations may have full bike stands, limiting the ability to return bikes.
- Limited availability of banking and bonus facilities at stations may reduce service convenience.
- Differences in station capacity (small, medium, large) may impact bike availability and system efficiency.
- Lack of clear understanding of bike availability trends over time may affect operational planning.
- Variation in average available bikes and bike stands between cities may indicate imbalance in resource allocation.

Tools and Technologies

- **Excel & Power Query** – Data cleaning and transformation
- **DAX (Data Analysis Expressions)** – Calculations and measures
- **Power BI** – Data visualization and dashboard development

Key Insights

- The network contains **thousands of bike stands distributed across multiple cities**.
- **Station capacity varies significantly**, with some stations categorized as small, medium, or large.
- Some stations have **zero available bikes**, indicating high usage or demand.
- Larger cities generally show **higher average bike and stand availability**.
- Only a **small number of stations provide banking and bonus services**, indicating limited premium features.

Conclusion

The dashboard provides insights into **bike availability, station capacity, and city-level distribution**. By analyzing capacity categories, availability levels, and geographic distribution, stakeholders can identify opportunities to **optimize bike allocation, improve station utilization, and enhance service accessibility**.

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